

Predicting post-pandemic CO² emissions in major European countries.

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I. INTRODUCTION

In 2020, the COVID-19 pandemic halted industrial and transportation activities across Europe, slashing CO² emissions by up to 12% in countries like Spain, France, and Italy [1], offering a rare glimpse into a low-carbon future. One of the most significant environmental changes observed was a noticeable drop in greenhouse gas emissions, particularly CO², due to reduced human activity. Predicting CO² emissions is essential for Europe, a region with significant industrial output and a global leader in climate action, to meet ambitious targets like the EU's 2050 net-zero goal [5]. However, predicting emissions of European countries is a complex challenge due to fluctuating economic recovery patterns, evolving policy frameworks, and diverse energy consumption behaviours across countries.

This crisis provided an opportunity to examine the relationship between human activity and environmental impact. By analyzing CO² emissions trends during lockdowns and reopening phases, this study identified factors driving emissions changes. The complexity came from integrating heterogeneous data such as a countries stringency index, GDP, and policy impacts, which vary unpredictably across nations and recovery phases. As shown in Figure 1, these factors significantly complicate accurate predictions.

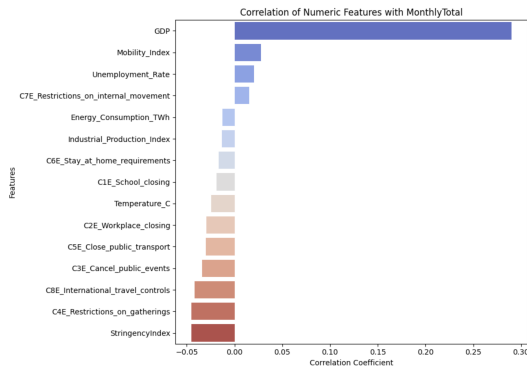


Fig. 1. CO² emissions feature importance

Given Europe's significant contribution to global emissions and its leadership in sustainability, studying CO² emissions in countries such as Italy, France, and Spain is particularly relevant. This study integrates findings from analyses of CO² emissions across major European countries during the COVID-19 pandemic, evaluating methodologies, identifying key influencing factors, and exploring their implications for predictive modelling.

II. RELATED RESEARCH

The COVID-19 pandemic sparked a wave of studies examining its impact on the environment, particularly on greenhouse gas emissions [8]. With widespread lockdowns and restrictions, human activity drastically declined, providing researchers with a unique chance to observe its effects on CO² levels. In response, a growing body of research has focused on both the short-term drops and the longer-term implications for emissions. This section reviews current studies on CO² emissions forecasting in the post-pandemic period, highlighting machine learning approaches, the role of different datasets, and the influence of economic and policy measures.

A. Predictive Modelling Techniques for CO₂ Emissions

Recent studies demonstrated a range of techniques for forecasting CO² emissions. Deng [4] applied machine learning (ML) models, including XGBoost and neural networks, to estimate fossil fuel-based CO² emissions in G20 nations after the pandemic. Their results showed machine learning's ability to capture complex, non-linear patterns better than traditional models.

Similar conclusions were reached by Mohan [10], who employed supervised machine learning models to assess COVID-19's broader impact, emphasizing the robustness and accuracy of machine learning during rapidly changing conditions. Bakay and Ağbulut [2] also used deep learning models to forecast greenhouse gas emissions from Turkey's power sector, reinforcing confidence in AI-based forecasting tools.

B. Data Sources for Emissions Forecasting

Robust emissions forecasting depends heavily on high-quality data. National Inventory Reports (NIRs) and the European Environment Agency’s Greenhouse Gas Emissions Database offer reliable country-level figures. Google Mobility Reports proved helpful during the pandemic, giving insight into population movement trends. Complementary data came from the Global Carbon Project and the Oxford COVID-19 Government Response Tracker (OxCGRT), both of which captured emissions shifts and policy stringency levels. In urban contexts, Eddy-covariance datasets have proven valuable for detecting shifts in carbon fluxes. Studies such as Le Quéré [9] and Forster [6] used this data to analyze how city-level emissions changed throughout lockdowns, highlighting how behavioral shifts influence local carbon cycles.

C. Policy and Economic Influences on Emissions Predictions

Policy and economic decisions play a major role in shaping future emissions. Rochedo [11] used Integrated Assessment Models (IAMs) to project reductions of 6% to 15% under various green recovery plans. These results point to the weight policy choices hold in meeting climate goals.

Likewise, Shan [3] applied adaptive regional input-output (ARIO) models and found emissions from key economic sectors temporarily fell by around 4%–6% between 2020 and 2024. Their analysis reinforces the idea that temporary declines are not enough—lasting reductions require broader structural reforms.

D. Summary of Current Literature Review

In summary, current literature supports the growing importance of machine learning in post-pandemic CO₂ forecasting. Machine learning models consistently outperform traditional techniques and offer more adaptable forecasting tools. Data quality and accessibility remain central, and the effectiveness of emissions forecasts often hinges on the availability of comprehensive and timely data. Studies also show that policy responses and economic strategies greatly influence emissions trajectories. However, despite these developments, there remains a gap in applying these methods at local levels.

E. Gaps in Predictive Research

While research into emissions forecasting has advanced, key gaps remain. Deng [4] pointed out that most models still lack access to wider datasets including carbon tax rates, foreign investment trends, and energy efficiency metrics. Furthermore, many machine learning models would benefit from deeper training datasets and the use of more advanced architectures like LSTM and CNNs to improve long-term forecasting.

There’s also a need for more localized predictions. Few models specifically focus on city-level emissions in

Europe, limiting insights for urban climate planning. That said, this study opted to focus on country-level emissions due to the availability of more complete and standardized national datasets. This approach ensures greater model reliability and allows for comparisons across major European countries.

III. METHODOLOGY

This section outlines the methodology and data preprocessing steps for forecasting monthly CO₂ emissions for European countries using three models in our first dataset: ARIMA, SARIMAX, XGBoost and in our second dataset: SARIMA, Scaled Wavelet-SARIMA, ETS and Prophet. The datasets, contained monthly CO₂ emissions and exogenous variables, covering the period including the COVID-19 years (2020–2023). Models were trained on data up to January 2023, tested from February 2023 onward, and forecasted to January 2026.

A. Data Description

Our first dataset included monthly CO₂ emissions (MonthlyTotal) and exogenous variables (GDP, Energy Consumption, Stringency Index, Industrial Production Index, Temperature, Mobility Index and Unemployment Rate) for multiple countries, with a Month column (YYYYMM format) and a Name column for country identification. The data captures emission trends, including disruptions during the COVID-19 period (2020–2023), influenced by lockdowns and policy changes.

Name	Substance	Month	StringencyIndex	MonthlyTotal	GDP	CHE_School	chse	Workplace	CHE_Cancel	public events	CHE_Restrictions	engart	CHE_Close	public tr
Austria	CO2	2020-01	0	7164.131444	709090118	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-02	0	6757.727761	709090934	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-03	0	6331.441468	709090959	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-04	0	5235.928319	704277901	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-05	0	4891.327437	704277901	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-06	0	4608.869008	709043016	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-07	0	4361.912000	709090929	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-08	0	4146.105641	713356332	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-09	0	4855.074430	713356332	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-10	0	6348.541798	709040464	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-11	0	6380.914029	704277915	0	0	0	0	0	0	0	0	0
Austria	CO2	2020-12	0	6998.552768	713356332	0	0	0	0	0	0	0	0	0

Fig. 2. CO₂ emissions combined with Stringency Index and exogenous variables dataset

B. Data Preprocessing

Data preprocessing was performed using Python with libraries such as pandas, numpy, and statsmodels. The steps, applied consistently across models unless specified, are as follows:

1) **Data Loading:** Our dataset was loaded, with column names standardized (spaces replaced with underscores).

2) **Country Filtering:** Data was filtered by country (Name) to create country-specific time series for each of the 10 countries.

3) **Time Index Setup:** The Month column was converted to a datetime format (YYYY-MM) and set as the index, ensuring a consistent monthly frequency.

4) **Missing Value Handling:** Rows with missing MonthlyTotal values were removed to ensure data integrity. For SARIMAX and XGBoost, missing exogenous variable values were imputed using forward-fill or mean imputation where necessary.

5) **Stationarity Check:** For ARIMA and SARIMAX, the Augmented Dickey-Fuller (ADF) test was applied to assess stationarity of the CO₂ emissions series. Non-stationary series were differenced (first-order and seasonal, period=12) to achieve stationarity, as confirmed by ADF p-values below 0.05.

6) **Train-Test Split:** The data was split at January 2023, with data up to this point used for training and post-January 2023 for testing.

7) **Feature Engineering (XGBoost):** : For XGBoost, additional features were derived, including lagged CO₂ emissions, time-based features (e.g., month, year), and policy indicators (e.g., lockdown status), to capture temporal and contextual patterns.

C. Model Methodologies

1) **ARIMA:** The AutoRegressive Integrated Moving Average (ARIMA) model was applied to capture temporal patterns in CO₂ emissions without exogenous variables. The model used fixed parameters: non-seasonal order (p,d,q)=(1,1,1) and seasonal order (P,D,Q,m)=(1,1,1,12), with a 12-month seasonal period to account for annual cycles. No trend component was included (trend="n"). The model was fit using the statsmodels library, with forecasts generated for the test period and extended to January 2026.

2) **SARIMAX:** The Seasonal ARIMA (SARIMAX) with Exogenous variables extended ARIMA by including exogenous variables (GDP, Energy Consumption, Stringency Index, Industrial Production Index, Temperature, Mobility Index and Unemployment Rate) to model external influences, particularly relevant during the COVID-19 period. The same fixed parameters as ARIMA (order=(1,1,1), seasonal_order=(1,1,1,12)) were used, with exogenous variables standardized to ensure numerical stability. The model was fit using statsmodels.

3) **XGBoost:** The eXtreme Gradient Boosting (XGBoost) model treated the forecasting task as a regression problem, using a feature set including lagged CO₂ emissions, exogenous variables (GDP, Energy Consumption, Stringency Index, Mobility Index), and additional features (e.g., month, year, policy indicators). Hyperparameters were tuned via grid search (e.g., learning rate, max depth, number of estimators) to optimize performance. The xgboost library was used, with models trained per country.

D. Evaluation Metrics

Model performance was evaluated using:

1) **RMSE and MAE:** Absolute error metrics to assess prediction accuracy in kilotons.

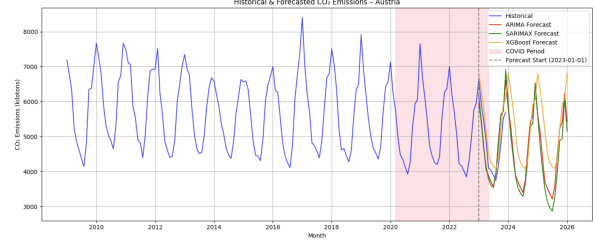


Fig. 3. Future Forecasting visual comparison of all 3 models on Austria

2) **MAPE:** Relative error metric (percentage) for scale-invariant accuracy.

3) **R-Squared:** Measure of model fit, indicating the proportion of variance explained.

Results were compared across the test period (February 2023 onward), with forecasts evaluated up to January 2026.

E. Carbon Monitor Dataset

The dataset utilised primarily for the predictive models trained pre-COVID emissions & validated on post-COVID emissions was the Carbon Monitor dataset. The carbon monitor dataset originates from an international initiative to regularly document CO₂ emission estimates while also revealing the decrease & increase of emissions during the pandemic. It contains all European countries, their daily emissions from 2019 - 2025, each emission-producing sector & a column called "Unnamed:" with null values. The carbon monitor dataset will allow our model to deploy a more detailed investigation of any trends or seasonality present in each country's emissions.

	A	B	C	D
1	country	date	sector	value
2	Norway	1/1/2019	Power	0.006542
3	Norway	2/1/2019	Power	0.005956
4	Norway	3/1/2019	Power	0.006048
5	Norway	4/1/2019	Power	0.00594
6	Norway	5/1/2019	Power	0.005649
7	Norway	6/1/2019	Power	0.005425
8	Norway	7/1/2019	Power	0.005163
9	Norway	8/1/2019	Power	0.004725
10	Norway	9/1/2019	Power	0.004504

Fig. 4. Carbon Monitor Dataset

To prepare the dataset for modeling, the "Unnamed: 4" column was dropped because it contained no values. Additionally, the title of the column is too vague, which means that we cannot use any of the other values in the dataset to try to determine what the original value was. The date column, which originally

contained the date values as strings, was converted to Datetime values & enforced the "Day, Month, Year" format.

Once these values had been successfully converted, the data was filtered to retain only the specified country. The final step involved aggregating the daily data to monthly data to create a new feature called "month_year". As the data contained daily emissions, this meant that it was harder to read the graphs constructed. With the monthly data, less noise is present, making it ideal for identifying large-scale shifts and seasonality.

F. Krushal-Wallis Test

To begin the process of selecting a model, two tests were carried out. The first test is the non-parametric test, Kruskal-Wallis. This test was used to detect if there is any seasonality amongst the various emissions. Despite the range of emissions differing from each other, the results are always 73.0 for the Kruskal-Wallis H-statistic & 0.478 for the p-value. Based on these results, there is no seasonality detected amongst the countries' emissions.

This is likely due to the grouping, which is based on the "month_year" feature. This means the test compares the distribution of CO² emissions for January across all years in the dataset, then the distribution for February across all years, and so on, within the filtered data for the selected country.

The general pattern of CO² emissions throughout the years (e.g., higher in winter, lower in summer) is similar across the countries tested. Even if the actual emission values differ, the Kruskal-Wallis test may not detect a significant difference in the distribution of these monthly values across countries. It's testing whether the shape and spread of the monthly data differ, not necessarily the overall average level.

G. ADF Tests

To check if the data for each country was stationary, we used the Augmented Dickey-Fuller test. When comparing the ADF results of the countries with the best, average, and worst overall models (Croatia, Austria, and Germany, respectively), it is evident that without differencing, Croatia and Germany are non-stationary. Croatia reported an ADF statistic of -1.1493 with a p-value of 0.5736 and Germany an ADF statistic of -1.7235 with a p-value of 0.4191. These results indicate that the p-value is greater than the typical significance level (0.05) and the ADF statistic is not more negative than the critical values; as a result, we have failed to reject the null hypothesis of a unit root. Based on the results provided, SARIMA was selected as the initial model for the dataset. Austria is already stationary with an ADF statistic of -6.0422 with a p-value of 0.0000.

Once differenced, we gain more insight into the potential d parameter. Croatia and Austria return ADF statistics of -3.2582 and -6.3645, with p-values of 0.0000 for both. We can determine from these results that the data is stationary and will most likely only need to be differenced once for Croatia, as it is well below the significance level, with the ADF statistic likely being more negative than the critical value. Since Austria is already stationary, no differencing will be required, so the d = 0.

For Germany, based on its results for the ADF Statistic being -2.7856 and a p-value of 0.0604, we have still failed to reject the null hypothesis at the 5% level. The differenced series is borderline non-stationary at 5% significance level; however, by differencing it again, it can be considered stationary at a 10% significance level.

H. ACF & PACF

The final process for model selection is the ACF & PACF. All three countries (Croatia, Austria, and Germany) exhibit very similar patterns in their ACF and PACF plots, suggesting that the underlying time series data likely shares similar characteristics or is influenced by similar factors. In all ACF plots, the autocorrelation values decay slowly over time, rather than dropping off sharply after a few lags. This slow decay is a strong indicator of non-stationarity in the time series. A stationary series would typically show correlations quickly dropping to zero.

Many lags, especially at regular intervals (e.g., around 12, 24, 36), show significant positive or negative correlations, extending beyond the confidence interval. This suggests the presence of seasonality. The consistent spikes around every 12th lag are particularly indicative of a yearly seasonality, given that the data is monthly. There's an alternating pattern of positive and negative correlations, which often points to a cyclical or seasonal component in the data.

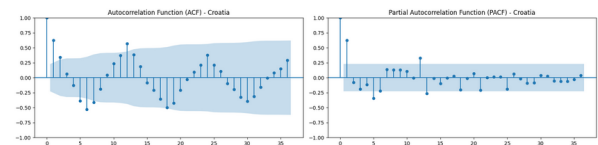


Fig. 5. Croatia ACF & PACF

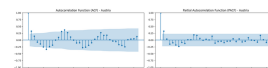


Fig. 6. Austria ACF & PACF



Fig. 7. Germany ACF & PACF

In all PACF plots, there's a significant spike at lag 1 (and often a significant negative spike at lag 2), followed by a rapid drop-off in most subsequent lags within the confidence interval, except for some specific

lags. Similar to the ACF, there are noticeable spikes at seasonal lags (e.g., around lag 12, and potentially lag 24), although these are generally less pronounced than in the ACF. The significance of these seasonal lags in the PACF further supports the presence of a strong seasonal component.

While the ACF shows a slow decay, the PACF generally shows a more rapid decay after the initial few significant lags, with most subsequent lags falling within the confidence interval, except for the seasonal spikes. Based on these observations, we can see that the data is non-stationary due to the slow decay, which is supported by the ADF results. Seasonality is present due to the significant spikes at regular intervals in the ACF and PACF. Therefore, a seasonal ARIMA (SARIMAX) model with seasonal differencing would be the most appropriate choice for our data.

I. Hyperparameter Tuning

To get the optimal parameters for the SARIMAX and Wavelet-SARIMA models, hyperparameter tuning is required. A grid of potential values for the non-seasonal parameters (p , d , q) and the seasonal parameters (P , D , Q) of the SARIMAX model, ranging between 0 and 2, was constructed. It then loops through every possible combination of these defined parameters. For each combination, it attempts to fit a SARIMAX model to the training data. Error handling is included to catch combinations that might fail to converge. After fitting, the code calculates several evaluation metrics for each successful model fit. These metrics include AIC, BIC, RMSE, MAE, MAPE and R-squared. The models are then ranked based on each metric, calculating a combined rank across all metrics to identify the overall "best" performing model based on the grid search. The parameters and metrics of the best model are printed.

J. Prophet and ETS

Prophet and Exponential Smoothing (ETS) models were employed for forecasting Germany's carbon emissions, as they are well-suited for time series data with discernible trends and seasonality. Carbon emissions frequently exhibit predictable seasonal patterns—for instance, elevated levels of CO₂ Emissions during colder months due to heating demands or warmer periods from increased cooling energy consumption, which both Prophet and ETS are designed to model effectively. Prophet is particularly advantageous for its resilience to missing data, outliers, and abrupt trend shifts, making it highly suitable for real-world emission datasets that are often noisy or incomplete. Conversely, ETS provides a robust statistical framework for modeling time series with exponential trends, offering interpretable components that encompass error, trend, and seasonality. Both models were trained on data prior to the official end of the pandemic,

which occurred in May 2023, and validated using post-COVID data. [7]

K. Evaluation Metrics

Model performance was evaluated using:

1) **AIC and BIC**: Both criteria evaluate the balance between model fit and complexity, penalizing excessive numbers of parameters to prevent overfitting.

2) **RMSE and MAE**: Absolute error metrics to assess prediction accuracy.

3) **MAPE**: Relative error metric (percentage) for scale-invariant accuracy.

4) **R-Squared**: Measure of model fit, indicating the proportion of variance explained.

Results were compared across the test period (May 2023 onward), with forecasts evaluated up to April 2025.

IV. RESULTS & ANALYSIS

This section presents the results of our models, ARIMA, SARIMAX and XGBoost for our first CO₂ emissions dataset and SARIMAX, ETS and Prophet developed to forecast monthly CO₂ emissions for European countries from our datasets. The models were trained on data up to January 2023 and tested on data from February 2023 onward, with forecasts extending to 2026. Performance was evaluated using 6 metrics: AIC (Akaike Information Criterion) and BIC (Bayesian Information Criterion) to compare the goodness of fit of statistical models, Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) for absolute accuracy, Mean Absolute Percentage Error (MAPE) for relative accuracy, and R-squared for model fit. The results compare model performance, identify the best model for each country, and discuss implications for CO₂ emissions forecasting.

A. Results of CO₂ emissions Models

This was our first dataset we attempted, the performance metrics for each model across the 10 countries are presented in Table I. MAPE and R-squared are prioritized for comparison due to their scale-invariant properties, while RMSE and MAE provide context for absolute errors, which vary with each country's emission magnitude. MAPE values are reported as percentages, and R-squared values are rounded to four decimal places for consistency.

The average performance across the 10 countries: ARIMA: MAPE = 2.92%, R-squared = 0.9356, SARIMAX: MAPE = 4.34%, R-squared = 0.8516 and XGBoost: MAPE = 9.02%, R-squared = 0.4529. ARIMA achieves the lowest MAPE in 8 countries (Austria, Germany, France, Ireland, Netherlands, Poland, Spain, Sweden) and highest R-squared in 7, excelling for small (Sweden: MAPE 1.85%, R-squared 0.9717) and large (Germany: MAPE 1.53%, R-squared 0.9861)

emitters. It captured temporal patterns effectively but underperformed in Belgium (MAPE 7.81%, R-squared 0.7891).

SARIMAX outperformed ARIMA in Belgium (MAPE 4.73% vs. 7.81%) and Italy (MAPE 2.19% vs. 2.34%) due to exogenous variables (GDP, Energy Consumption, Stringency Index, Mobility Index), but has higher MAPE and lower R-squared in 8 countries, notably Ireland (R-squared 0.5841) and Spain (R-squared 0.7133).

XGBoost consistently underperformed (highest MAPE, lowest R-squared), with negative R-squared in Ireland (-0.0735) and Poland (-0.1311), due to its non-time-series approach.

ARIMA was the most effective model, capturing temporal patterns in CO₂ emissions with high accuracy. SARIMAX's exogenous variables added value in Belgium and Italy, while XGBoost was unsuitable without time-series enhancements. The COVID period's volatility is well-handled by ARIMA's differencing and SARIMAX's variables. The weaker performance in Ireland and Spain suggest further investigation into the exogenous variables to improve model fit. These results support reliable short-term CO₂ forecasting, with the second dataset promising further improvements through enhanced data and variables.

B. Results of Carbon Monitor Models

The first model used was SARIMAX. When the results of all countries were averaged, it achieved an RMSE of 0.8165, an MAE of 0.6698, an MAPE of 6.3885, and an R-squared of 0.6372. The RMSE and MAE are relatively close, suggesting that there aren't many huge errors skewing the RMSE significantly more than the MAE. The MAPE of 6.39% indicates that the predictions are, on average, within a reasonable percentage of the actual values. The R-squared value of 0.6372 indicates that the model explains a substantial portion of the variability in the data. Although it's not exceptionally high compared to the R-squared achieved, it indicates that the model has predictive power. In 10 out of the 15 countries, the R-squared is above 0.7400, with France, Luxembourg, and Poland entering the 0.9000 threshold.

Among the four models created, the scaled Wavelet-SARIMA provided the most accurate prediction when compared to the actual post-COVID emissions. When the model was trained with Denmark's CO₂ emissions, we can see that the values given for AIC & BIC are quite low at 14.58 & 13.02. Indicating that the model achieves a good balance between complexity & model fit.

With an RMSE: 0.13, MAE: 0.12 & MAPE: 4.29, we can see that these evaluation metrics are all relatively low. The RMSE & MAE, which are low, show that

the model has few predictive errors. Although MAPE indicates that, on average, the predictions are off by 4% from the actual values. The R-squared of 0.81 further emphasises that the model explains approximately 81% of the variance in the data. After calculating the averages of the models, we get a value of 1.6786 for the RMSE, confirming that the Wavelet-SARIMA model outperformed the other models.

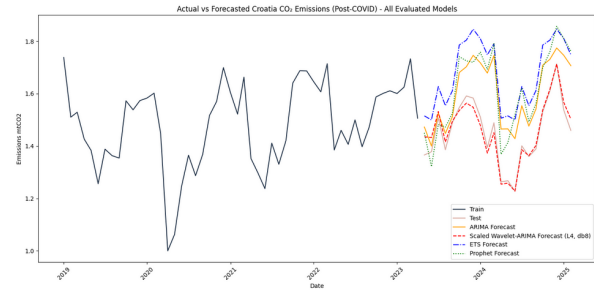


Fig. 8. Best Predictive Performance



Fig. 9. Average Predictive Performance

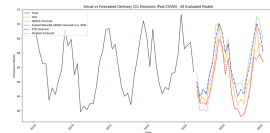


Fig. 10. Worst Predictive Performance

The ETS and prophet model, while providing good results in terms of RMSE, MAE, and MAPE, with an average of the results for the former being RMSE: 1.4699, MAE: 1.3270, MAPE: 15.5005, and for the latter RMSE: 1.7072, MAE: 1.4718, MAPE: 15.4371. They had relatively low R-squared scores across all countries, with the three highest averaged values being 0.77 for Poland, 0.06322 for Luxembourg, and 0.7231 for Hungary. Ireland displays the lowest R-squared score with a value of -58.6132. On average, the R-squared value for the ETS model was -3.7653 and -4.0144 for Prophet, indicating that Prophet was the worst-performing model among the four used, with ETS performing slightly better. It can be concluded that they provided a poor fit for the data and struggled to capture the underlying patterns.

When compared to the other three models, the SW-SARIMA very accurately predicts the actual values of post-COVID emissions. Upon inspecting the graph, the other models predicted higher emissions than the actual data. This is likely because wavelet decomposition separates trends, seasonality, and noise. Since the original time series is decomposed into various frequency components, wavelets allow SARIMA to model the long-term trends cleanly. [12] This leaves the detailed components to be modeled individually, allowing for better identification of underlying patterns.

V. CONCLUSION & FUTURE WORK

This study examined the impact of the COVID-19 pandemic on CO₂ emissions across major European countries and evaluated multiple forecasting models to predict post-pandemic trends. Through two comprehensive datasets—the combined exogenous dataset and the Carbon Monitor dataset—we applied a suite of models including ARIMA, SARIMAX, XGBoost, ETS, Prophet, and Scaled Wavelet-SARIMA to assess model effectiveness in capturing CO₂ dynamics.

From the first dataset, ARIMA emerged as the most consistently accurate model across countries, effectively handling time-series characteristics of emissions. SARIMAX proved beneficial in specific contexts (e.g., Belgium and Italy), leveraging exogenous variables to improve forecasts. XGBoost underperformed due to its lack of temporal structure and weaker adaptation to time-dependent patterns.

In the Carbon Monitor dataset, Scaled Wavelet-SARIMA outperformed all other models, achieving the lowest RMSE and MAPE while maintaining strong R-squared values. The model's strength lies in its ability to decompose time series into trend, seasonal, and noise components, thereby improving interpretability and forecast accuracy. Conversely, Prophet and ETS exhibited poor performance, especially in capturing the long-term structure and variance of emissions data.

These findings highlight that no single model fits all cases equally rather, model performance varies based on data structure, feature quality, and national emission behaviours. Models incorporating time-series decomposition and differencing (e.g., ARIMA and SW-SARIMA) consistently deliver stronger forecasts, reinforcing the importance of temporal modelling in environmental data.

A. Future Work

Several opportunities exist for extending this research:

- **City-Level Forecasting:** While national-level emissions provide useful insights, modeling city-specific emissions could help local policymakers better plan urban climate strategies.
- **Incorporating Additional Features:** Future studies could incorporate carbon pricing data, investment in green technologies, and sector-specific emissions to enhance model granularity.
- **Deep Learning Approaches:** Architectures such as Long Short-Term Memory (LSTM) and Temporal Convolutional Networks (TCNs) could be explored for improved performance, especially in capturing long-term dependencies.
- **Hybrid and Ensemble Models:** Combining classical statistical models with machine learning (e.g., ARIMA-LSTM hybrids) may yield better generalization and predictive accuracy.

Overall, the results highlight the importance of model selection, data preprocessing, and feature engineering in forecasting CO₂ emissions.

TABLE I
CO₂ AND EXOGENOUS EMISSIONS RESULTS

Country	Model	RMSE	MAE	MAPE	R-Squared
Austria	ARIMA	128.67	109.20	2.53	0.9714
	SARIMAX	282.78	250.96	5.78	0.8619
	XGBoost	397.03	356.96	7.43	0.7278
Belgium	ARIMA	592.07	536.69	7.81	0.7891
	SARIMAX	346.19	304.35	4.73	0.9279
	XGBoost	662.06	618.40	9.07	0.7363
Germany	ARIMA	951.25	751.55	1.53	0.9861
	SARIMAX	2279.14	1884.23	3.87	0.9203
	XGBoost	7228.97	7028.30	15.10	0.1985
France	ARIMA	978.56	930.21	4.11	0.9410
	SARIMAX	1443.04	1153.12	5.56	0.8717
	XGBoost	2606.11	2423.88	10.69	0.5816
Ireland	ARIMA	101.51	82.94	2.99	0.8540
	SARIMAX	171.31	152.81	5.89	0.5841
	XGBoost	275.22	250.74	9.34	-0.0735
Italy	ARIMA	639.53	545.82	2.34	0.9636
	SARIMAX	669.95	553.57	2.19	0.9601
	XGBoost	1808.93	1598.06	6.25	0.7091
Netherlands	ARIMA	303.65	265.38	2.90	0.9656
	SARIMAX	498.40	369.13	4.24	0.9073
	XGBoost	1236.90	1184.55	12.23	0.4290
Poland	ARIMA	409.42	343.53	1.42	0.9776
	SARIMAX	948.07	641.48	2.64	0.8797
	XGBoost	2906.96	2698.55	11.35	-0.1311
Spain	ARIMA	387.74	318.44	1.71	0.9355
	SARIMAX	817.70	763.45	4.32	0.7133
	XGBoost	1157.61	947.55	5.56	0.4254
Sweden	ARIMA	62.14	50.87	1.85	0.9717
	SARIMAX	122.42	116.71	4.21	0.8900
	XGBoost	100.61	91.53	3.21	0.9257

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TABLE II
CARBON MONITOR RESULTS

Country	Model	AIC	BIC	RMSE	MAE	MAPE	R-Squared
Austria	SARIMAX	95.8065	99.1843	1.3015	0.8480	12.9764	-0.1969
	Scaled Wavelet-SARIMA	23.8683	23.4518	0.2205	0.1532	2.6449	0.9656
	ETS	-49.7129	-22.3955	1.4054	0.8605	13.0759	-0.3955
	Prophet	N/A	N/A	1.4421	0.9399	14.4883	-0.4693
Belgium	SARIMAX	97.0145	100.3417	0.5047	0.4020	5.1442	0.7438
	Scaled Wavelet-SARIMA	27.3949	26.2233	0.1807	0.1355	1.7842	0.9671
	ETS	-50.6675	-23.3501	0.7208	0.6239	8.6438	0.4775
	Prophet	N/A	N/A	0.9328	0.8122	11.1384	0.1247
Denmark	SARIMAX	3.9271	14.0604	0.1403	0.1166	4.0797	0.7750
	Scaled Wavelet-SARIMA	14.5845	13.0223	0.1284	0.1159	4.2852	0.8116
	ETS	-163.5403	-136.2229	0.1347	0.1129	3.9612	0.7925
	Prophet	N/A	N/A	0.2337	0.1826	6.4786	0.3753
France	SARIMAX	188.9359	195.6914	0.9209	0.7531	3.0880	0.9330
	Scaled Wavelet-SARIMA	37.5945	36.8134	0.5458	0.4028	1.6706	0.9764
	ETS	84.6229	111.9403	2.3073	2.1697	9.0732	0.5791
	Prophet	N/A	N/A	2.7314	2.3623	9.7894	0.4101
Italy	SARIMAX	178.6594	183.5722	1.2736	0.9669	3.9196	0.8230
	Scaled Wavelet-SARIMA	37.1338	35.9621	0.5430	0.4266	1.7480	0.9678
	ETS	63.2341	90.5515	3.1103	2.7417	11.2851	-0.0557
	Prophet	N/A	N/A	4.3557	3.9897	16.4602	-1.0702
Germany	SARIMAX	246.5731	254.7610	4.0216	3.4989	7.0000	0.8109
	Scaled Wavelet-SARIMA	51.9516	50.3893	1.8710	1.3516	2.6829	0.9591
	ETS	157.2468	184.5642	6.7679	6.3731	13.5225	0.4644
	Prophet	N/A	N/A	8.0416	7.4861	15.7547	0.2437
Ireland	SARIMAX	1.0851	7.8406	0.1301	0.1048	3.2180	0.8123
	Scaled Wavelet-SARIMA	18.3482	17.1765	0.0764	0.0539	1.6062	0.9353
	ETS	-320.5155	-293.1981	2.3066	2.2958	70.4219	-58.0206
	Prophet	N/A	N/A	2.3181	2.3074	70.7841	-58.6132
Luxembourg	SARIMAX	-113.2791	-109.9520	0.0246	0.0215	2.3252	0.9158
	Scaled Wavelet-SARIMA	-0.7483	-1.9200	0.0335	0.0207	2.3896	0.8443
	ETS	-320.5155	-293.1981	0.0541	0.0509	5.5964	0.5947
	Prophet	N/A	N/A	0.0488	0.0436	4.7932	0.6696
Poland	SARIMAX	157.8788	164.5331	1.2177	0.9079	3.7524	0.9013
	Scaled Wavelet-SARIMA	42.0866	40.5244	0.7629	0.5315	2.3349	0.9613
	ETS	38.2180	65.5354	1.9722	1.7656	7.9189	0.7411
	Prophet	N/A	N/A	1.7376	1.5985	7.1918	0.7989
Estonia	SARIMAX	17.7420	26.0598	0.1706	0.1347	17.3144	-0.0405
	Scaled Wavelet-SARIMA	19.7799	18.2177	0.0352	0.0232	2.7810	0.9557
	ETS	-145.5450	-118.2276	0.2372	0.1850	24.5021	-1.0117
	Prophet	N/A	N/A	0.3269	0.2729	35.7876	-2.8236
Netherlands	SARIMAX	123.1037	131.4215	1.3497	1.2709	11.9015	0.3959
	Scaled Wavelet-SARIMA	35.7803	34.2180	0.4942	0.3931	3.5379	0.9190
	ETS	-4.8962	22.4212	1.4100	1.2898	12.1598	0.3407
	Prophet	N/A	N/A	1.5665	1.3967	13.1032	0.1861
Greece	SARIMAX	73.8970	77.2241	0.3348	0.2701	5.1685	0.4710
	Scaled Wavelet-SARIMA	23.6673	22.1050	0.3125	0.2571	4.8450	0.5392
	ETS	-72.5837	-45.2663	0.3518	0.2909	5.4152	0.4161
	Prophet	N/A	N/A	0.3778	0.3193	6.1717	0.3262
Hungary	SARIMAX	-0.0058	4.9849	0.1744	0.1414	3.9253	0.8619
	Scaled Wavelet-SARIMA	18.9897	17.8180	0.1672	0.1164	3.3446	0.8731
	ETS	-164.6709	-137.3534	0.2729	0.2380	6.8628	0.6617
	Prophet	N/A	N/A	0.2178	0.1716	4.9422	0.7844
Croatia	SARIMAX	-66.7195	-61.6529	0.0795	0.0642	4.4566	0.5615
	Scaled Wavelet-SARIMA	8.1309	6.5686	0.0283	0.0219	1.5153	0.9445
	ETS	-243.9311	-216.6137	0.2318	0.2226	15.4579	-2.7265
	Prophet	N/A	N/A	0.1889	0.1685	11.6792	-1.4781
Czech Republic	SARIMAX	79.3249	85.9791	0.6037	0.5472	7.5578	0.7910
	Scaled Wavelet-SARIMA	32.4171	30.8549	0.2818	0.2246	3.0738	0.9545
	ETS	-62.1625	-34.8451	0.7666	0.6858	9.6105	0.6630
	Prophet	N/A	N/A	1.0897	1.0261	14.2788	0.3190

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